American Journal of Epidemiology, 2025, **194**, 185–194
https://doi.org/10.1093/aje/kwae171
Advance access publication date July 3, 2024
Practice of Epidemiology

A causal machine-learning framework for studying policy impact on air pollution: a case study in COVID-19 lockdowns

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Abstract

When studying the impact of policy interventions or natural experiments on air pollution, such as new environmental policies or the opening or closing of an industrial facility, careful statistical analysis is needed to separate causal changes from other confounding factors. Using COVID-19 lockdowns as a case study, we present a comprehensive framework for estimating and validating causal changes from such perturbations. We propose using flexible machine learning–based comparative interrupted time series (CITS) models for estimating such a causal effect. We outline the assumptions required to identify causal effects, showing that many common methods rely on strong assumptions that are relaxed by machine learning models. For empirical validation, we also propose a simple diagnostic criterion, guarding against false effects in baseline years when there was no intervention. The framework is applied to study the impact of COVID-19 lockdowns on atmospheric nitrogen dioxide (NO_2) levels in the eastern United States. The machine learning approaches guard against false effects better than common methods and suggest decreases in NO_2 levels in 4 US cities (Boston, Massachusetts; New York, New York; Baltimore, Maryland; and Washington, DC) during the pandemic lockdowns. The study showcases the importance of our validation framework in selecting a suitable method and the utility of a machine learning–based CITS model for studying causal changes in air pollution time series.

This article is part of a Special Collection on Environmental Epidemiology.

Key words: machine learning; interrupted time series; causal inference; COVID-19; air pollution.

Introduction

In 2020, lockdowns and other restrictions were implemented worldwide due to the COVID-19 pandemic. This decreased traffic volume.[1] Traffic is a major source of certain pollutants, like nitrogen dioxide (NO_2).[2] Many studies have been published on the effects of COVID-19 lockdowns on ambient levels of air pollutants, including NO_2. Gkatzelis et al[3] reviewed over 200 such papers. A variety of statistical methods and analyses were used, including measurement of percent reduction[4-8] between the lockdown period and a reference period, comparison of concentrations in the same period over multiple years,[9,10] the difference-in-differences approach,[11,12] and regression models, including comparative interrupted time series (CITS)[13,14] or prediction of counterfactual concentrations.[15,16] Models of air pollution under different emission scenarios, including chemical transport models, were also used.[17-20]

These methods often suffer from some of the following limitations. First, meteorological conditions were not always considered in previous studies. COVID-19 lockdowns in the United States were generally imposed around March 2020, and many pollutant concentrations increase in winter and decrease in summer in parts of the Northern Hemisphere due to meteorological factors, including greater emissions from combustion-based heating sources during winter.[21,22] Not including weather conditions would confound the lockdown effect estimates.

Second, many researchers do not estimate a causal effect or rely on unstated and unrealistic assumptions. The COVID-19 lockdown constitutes an example of a "sharp" policy intervention with a potential impact on air pollution. Other examples of such policies or natural experiments include copper smelt strikes in the southwestern United States,[23,24] steel mill closures in the US state of Utah,[25] and emissions reductions during the 2008 Olympic Games in Beijing, China.[26] Conclusions about a decrease in air pollution have often been based on comparisons of average concentrations during and before the experiment. It is important to assess whether such common approaches provide a causal estimate.

Third, parametric forms such as linearity are often assumed, which, if inaccurate, would bias estimation despite the inclusion of meteorological data.

The contributions of this article are 2-fold. First, we propose a flexible machine learning–based CITS framework for studying

Received: May 14, 2023. **Accepted:** June 26, 2024
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causal changes in air pollution due to a sharp intervention. Machine learning has been used to predict "business as usual" concentrations,[16] and nonlinear parametric CITS models exist,[27] but our approach to using machine learning in CITS is novel. Because of the complexity of air pollution data, machine learning approaches are appealing for studying policy interventions, due to their minimal reliance on parametric assumptions. A nonparametric CITS could be used in settings other than air pollution, making this approach valuable in other contexts.

Second, we present a comprehensive framework for conceptual and empirical validation of any approach studying such causal changes. We outline the assumptions that allow identification of a causal effect of a sharp intervention. We show in the Methods section that many popular approaches used in such problems are special cases of CITS (see "Common methods" subsection) with strong and unrealistic assumptions (see "Conceptual validation"), making estimates from these methods less reliable. Machine learning methods yield a causal estimand under minimal assumptions, and allow complex nonlinear relationships between meteorology and air pollution to be modeled. We then present a broadly applicable empirical validation strategy (see "Empirical validation"): using a candidate method on data from baseline years without an intervention. Identification of causal changes in baseline years indicates a method's inadequacy. Such validation has not been done in most studies on the effects of sharp interventions on air pollution, so our framework serves as a guide to validating these causal effects.

We present a case study evaluating a suite of existing and proposed methods for identifying changes in NO_2 concentrations due to the COVID-19 lockdowns in 5 eastern US cities: Boston, Massachusetts; New York, New York; Philadelphia, Pennsylvania; Baltimore, Maryland; and Washington, DC.

Methods

Causal effect: average treatment effect on the treated

The causal estimand used to understand the impact of a policy intervention on air pollution is the average treatment effect on the treated (ATT). Here *treated* is the group after the intervention. In our case study, the ATT calculates the effect of the lockdown on time points in 2020 in which people experienced the lockdown.

Let Y be the pollutant concentration of interest and A be the indicator of the intervention (e.g., COVID-19 lockdowns), representing whether the intervention was in effect ($A = 1$) or not ($A = 0$) at time t. Let $Y^a(t)$ be the potential outcome of the concentration at time t given $A = a$ ($a \in \{0, 1\}$). Then the ATT at time t is

$$\mathbf{ATT}(t) = E\left[Y^1(t) - Y^0(t) | A = 1\right].$$

The average of $\mathbf{ATT}(t)$ over a set of time points T (which we refer to as the ATT) is

$$\mathbf{ATT} = E\left[\overline{Y^1}_T - \overline{Y^0}_T | A = 1\right], \quad (1)$$

where $\overline{Y^a}_T = \frac{1}{|T|}\sum_{t \in T} Y^a(t)$ and $|T|$denotes the total number of time points in the intervention period. In addition to reporting the ATT, we can calculate the percent difference in concentrations caused by the intervention:

$$\%\mathbf{diff} = \frac{E\left[\overline{Y^1_T} - \overline{Y^0_T} | A = 1\right]}{E\left[\overline{Y^0_T} | A = 1\right]} \times 100 = \frac{\mathbf{ATT}}{E\left[\overline{Y^0_T} | A = 1\right]} \times 100. \quad (2)$$

General CITS framework

We introduce a general CITS framework[28] that encompasses our proposed method and several commonly used methods. The ATT is estimated by modeling potential outcomes $Y^a(t)$, for times t before and after the intervention date in the intervention year, and at the same times t in previous years, or "baseline years." Here, t is the time stamp (hour-day-month) with no associated year. We use baseline years as a historical comparison group rather than a contemporary comparison group (more common in CITS), since an intervention such as COVID-19 lockdowns was implemented near-simultaneously in all regions of interest, so no contemporary control group exists.

Let P denote an indicator equaling 1 after the "cutoff" (date/time of the intervention) and 0 before the cutoff, and I equals 1 during the intervention year and 0 otherwise. We consider a set of meteorological covariates $\mathbf{Z}$ (see "Data and modeling" subsection). Note that $\mathbf{Z}$ cannot include mediator variables affected by the intervention, such as traffic, since conditioning on a mediator biases the intervention effect.[29] We propose a general model

$$E\left[Y^a(t)\right] = F(t, P, I, A = a, \mathbf{Z}) \quad (3)$$

and obtain the ATT using equation (1). This formulation generalizes CITS by using a general functional form F for $Y^a(t)$ instead of parametric forms.[28]

Machine learning CITS

We propose estimating F nonparametrically using any machine learning algorithm. In our case study, we consider gradient boosting (GB)[30] and random forest (RF).[31] Our proposed approach addresses the limitations mentioned in the Introduction. The model is nonparametric, thereby not restricting the relationship between the pollutant and other variables $\mathbf{Z}$, including meteorological variables. The method estimates the ATT under less restrictive assumptions than common methods (see "Conceptual validation").

Since many machine learning methods allow for arbitrary interactions among the variables, the set of covariates must not be able to recreate the intervention. For example, because $A = P \times I$, including both P and I can make the effect unidentifiable when using flexible models for F. Therefore, we omit I from equation (3).

Instead of using a single functional form F for the outcome model where the intervention A is a covariate ("S-learner"), one can estimate separate functional forms F_0 and F_1 for the potential outcomes ("T-learner"). We prefer the S-learner here, as there is much less data in the intervention period with which to estimate a separate function F_1, particularly if using machine learning. Using a single F allows borrowing information about meteorological effects on air pollution from baseline years, improving precision of the potential outcomes estimates in the intervention period.

Common methods

We show that several common estimation methods are special cases of the CITS framework. Let T_1 be the set of times t after the intervention and T_0 be the set of times t before the intervention.

Mean differences

1. Difference in mean concentrations between the postcutoff period and the precutoff period (DPPC) in the intervention year:[5,32]

$$\mathbf{DPPC}: \mathbf{ATT} = E\left[\overline{Y_{T_1}} | P = 1, I = 1\right] - E\left[\overline{Y_{T_0}} | P = 0, I = 1\right]. \quad (4)$$

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The ATT from the DPPC is equivalent to using CITS (equation (3)) with $E[Y^a(t)] = \beta_0 + \beta_1 A$ and estimating β_1 using data only from the intervention year (Appendix S1).

2. Difference in mean concentrations for the postcutoff period between the intervention year and the baseline years (DIB):[10]

$$\text{DIB} : \text{ATT} = E\left[\overline{Y_{T_1}} \mid P = 1, I = 1\right] - E\left[\overline{Y_{T_1}} \mid P = 1, I = 0\right]. \quad (5)$$

The ATT from the DIB is equivalent to using CITS (equation (3)) with $E[Y^a(t)] = \beta_0 + \beta_1 A$ and estimating β_1 using data only from the postintervention period in all years (Appendix S1).

3. Difference in differences (DiD): Consider (a) the difference in mean concentrations in the postcutoff period between the intervention and baseline years and (b) the difference in mean concentrations in the precutoff period between the intervention and baseline years. The DiD takes the difference between these two differences (a) and (b),[33] ie,

$$\text{DiD} : \text{ATT} = \left[E\left[\overline{Y_{T_1}} \mid P = 1, I = 1\right] - E\left[\overline{Y_{T_1}} \mid P = 1, I = 0\right]\right]$$

$$- \left[E\left[\overline{Y_{T_0}} \mid P = 0, I = 1\right] - E\left[\overline{Y_{T_0}} \mid P = 0, I = 0\right]\right]. \quad (6)$$

This quantity is the estimate of β_3 in a CITS (equation (3)) with $E[Y^a(t)] = \beta_0 + \beta_1 P + \beta_2 I + \beta_3 A$ (Appendix S1).

Thus, these common approaches can be expressed as special cases of CITS with linear regression on binary variables.

More general linear models have also been considered for CITS, as shown below.

Linear regression (with time splines) CITS

$$E[Y^a(t)] = \beta_0 + \beta_1 P + \beta_2 I + s_3(t) + s_4(t) \times P$$

$$+ \beta_5 t \times I + \text{IE}(A = a, t) + \beta_8{}'\mathbf{Z}, \quad (7)$$

where $s_i(t)$ are splines of time and where the intervention effect (IE), $\text{IE}(A, t) = \text{ATT}(t)$, can be either

4. constant (LC): $\text{IE}(A, t) = \beta_6 A$ or
5. time-varying (LTV): $\text{IE}(A, t) = \beta_6 A + s_7(t) \times A$.

These approaches explicitly use linear models with splines of time to allow for temporal structure.[34-36] They are more flexible than many CITS models that model time linearly and can capture complex seasonal variations in air pollution concentrations. All of these approaches estimate the ATT under certain assumptions (see "Conceptual validation"), but the assumptions are not always feasible. The methods are summarized in Table 1.

Conceptual validation: assumptions for identifying the ATT

We present a validation framework for any method estimating a sharp intervention's effect on air pollution. First, we present conceptual validation, considering the assumptions under which these approaches estimate the ATT and determining whether these are reasonable. We must assume conditional ignorability, that no change affecting air pollution during the lockdowns that is not due to the lockdowns occurred; otherwise we cannot disentangle the effects of 2 simultaneous interventions. Formally, the assumptions made in order to estimate the ATT[33,37] are

(1) consistency,
(2) conditional ignorability,
(3) positivity in the treated population, and
(4) parallel trends.

In Appendix S2, we provide technical definitions of these conditions and a discussion interpreting these assumptions, for a general intervention and specifically for COVID-19 lockdowns. These assumptions provide a validation framework for conceptually assessing any method used to obtain a causal effect of a sharp policy change.

For machine learning CITS, the assumptions above are sufficient to estimate the ATT. However, linear CITS requires another assumption:

(*) The linear model form is correctly specified.

Appendices S3 and S4 show that with this assumption, the "parallel trends" assumption is satisfied and the ATT is identifiable. Thus, these models require the strong assumption of a linear model between concentrations and covariates.

The mean difference methods shown above in the "Common methods" subsection do not use any covariates $\mathbf{Z}$, making conditional ignorability much stronger but positivity and parallel trends trivial. Each method makes another assumption:

(**) DPPC: The mean concentration for the intervention year precutoff period is the same as the mean of the counterfactual concentration for the postcutoff period with no intervention (Lemma 1).

(***) DIB: The mean counterfactual concentration for the intervention year postcutoff period with no intervention is equal to the mean concentration for the same period in baseline years (Lemma 2).

(****) DiD: The mean counterfactual concentration for the intervention year postcutoff period with no intervention is equal to the mean concentration for the same period in baseline years plus a yearly effect equal to the difference in average concentrations in the precutoff period between the intervention and baseline years (Lemma 3).

The formal statements and proofs of the lemmas are shown in Appendix S5. If these assumptions are justified, each method yields a causal estimate of the ATT. We also discuss the issues with making these assumptions in our case study, namely that differing meteorological conditions would make the assumptions questionable or inappropriate. In general, any application of these methods should assess whether the additional assumptions made are reasonable in a particular context.

Empirical validation

We propose an empirical validation approach that can be applied universally to any method assessing causal changes in air pollution due to a sharp intervention. We aim to ensure that a method does not produce a large nonzero ATT if applied to a year with no intervention. To check this, we apply the method with each baseline year as the year of interest. Details on setting up this validation in our case study are provided in Appendix S6. If a method results in near-zero ATT estimates for all baseline years, the result in the intervention year is unlikely to be a false effect. Additionally, if the proposed method offers formal statistical testing, a *P* value may also be calculated to assess the significance of effects in baseline years. Any effects with large magnitudes, such as a magnitude similar to the magnitude in 2020, or a significant *P* value in baseline years are undesirable.

In our case study of COVID-19 lockdowns, our mean difference methods (DPPC, DIB, DiD) are all equivalent to parametric

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Table 1. Models and abbreviations used in the current study

Abbreviation	Model name	Method description
DPPC	Difference between the mean values in the postcutoff and precutoff periods	The difference between the average concentrations before and after the intervention in the year the intervention took place. The formula is given in equation (4).
DIB	Difference between the mean values for the postcutoff period in the intervention year and the baseline years	The difference between the mean concentration in the dates after the intervention in the year of the intervention compared with baseline years. The formula is given in equation (5).
DiD	Difference in differences	Take the difference in the average concentration after the intervention in the intervention year and baseline years, as well as the difference in the average concentration in the dates before the intervention in the intervention year and baseline years. Then take the difference between these 2 differences. The formula is given in equation (6).
LC	Linear model with a constant lockdown effect	Fit a parametric model for the potential outcomes that is linear in the covariates (plus splines in time) where the effect of the intervention is constant over time. The formula is given in equation (7) with a constant intervention effect.
LTV	Linear model with a time-varying lockdown effect	Fit a parametric model for the potential outcomes that is linear in the covariates where the effect of the intervention varies over time. The formula is given in equation (7) with a time-dependent intervention effect.
GB	Gradient boosting	Fit a nonparametric model for the potential outcomes using gradient boosting. The formula is given in equation (3) without *I*. We use the XGBoost R package.
RF	Random forest	Fit a nonparametric model for the potential outcomes using random forest. The formula is given in equation (3) without *I*. We use the randomForestSRC R package.

specifications of the regression form *F*, as explained above in "Common methods." This means that the significance of the coefficient of the intervention indicator *A* assesses the significance of the ATT. For the linear model with a constant lockdown effect (LC), the significance of the intervention indicator term can also be used, since it appears only once in the model. For a linear model with a time-varying lockdown effect (LTV), an *F* test must be performed to assess the significance of all terms involving the intervention. For the machine learning CITS, there is no such method for assessing the statistical significance of the ATT. We use block bootstrapping to obtain interval estimates of the ATT.[38] The methods for assessing significance are explained further in Appendix S7 and Figures S1-S6, including a comparison of different bootstrapping approaches for machine learning CITS.

Data and modeling

For analysis of the COVID-19 lockdowns' impact on air quality, we consider NO_2 data from 5 cities in the northeastern United States: Boston, Massachusetts; New York, New York; Philadelphia, Pennsylvania; Baltimore, Maryland; and Washington, DC. NO_2 is our pollutant of interest because it is highly affected by traffic and many studies have found decreases in NO_2 levels due to COVID-19 lockdowns.[3] We use publicly available NO_2 data from the Environmental Protection Agency (EPA) website for the years 2013-2020. Use of these data ensures high-quality measurements, due to adherence to the EPA's strict standards. Thus, we can consider the data to represent the ground truth and do not need to further calibrate the measurements or otherwise correct for measurement inaccuracies.

Our covariates include relative humidity, temperature, barometric pressure, and wind speed from the EPA website. We also include wind direction, planetary boundary layer height, and precipitation rate, measured by the National Oceanic and Atmospheric Administration. We also use a weekday/weekend indicator, an hour-of-the-day variable, and an indicator for number of daylight hours, determined by the daily sunrise and sunset time in each city, since they are useful supplemental factors in predicting NO_2 concentrations.[39] Covariate values are averaged across a city, since covariate levels were often not recorded at the sites measuring NO_2, while NO_2 data are not averaged, and monitor-specific indicators are used as covariates. We do not include covariates related to NO_2 that could be affected by the lockdown (eg, traffic, ozone). There are some differences in the final datasets across cities (Appendix S8). The lockdown start (cutoff) for each city is the date on which a state of emergency was declared in that state, and each city is modeled independently. For the analysis, data from January 1 through 8 weeks past the city-specific lockdown date are used from each year to capture the period with the most adherence to lockdown restrictions (Appendix S8 and Figure S7). We consider the models reviewed above in the "Machine learning CITS" and "Common methods" subsections, summarized in Table 1. We compared 2 software packages to fit boosting models and selected XGBoost as our preferred boosting implementation (Appendix S9 and Figure S8).

Results

The time series of NO_2 concentrations in all cities in 2020 is compared with the time series of average concentrations across baseline years in Figure 1. In all cities, after the lockdown date, baseline concentrations are higher than the 2020 concentrations. Before the lockdown date, there is less difference between the time series. Time series for each year are included in Figure S9. The descriptive statistics for NO_2 and covariates and the number of measurement sites are shown in Tables S1 and S2.

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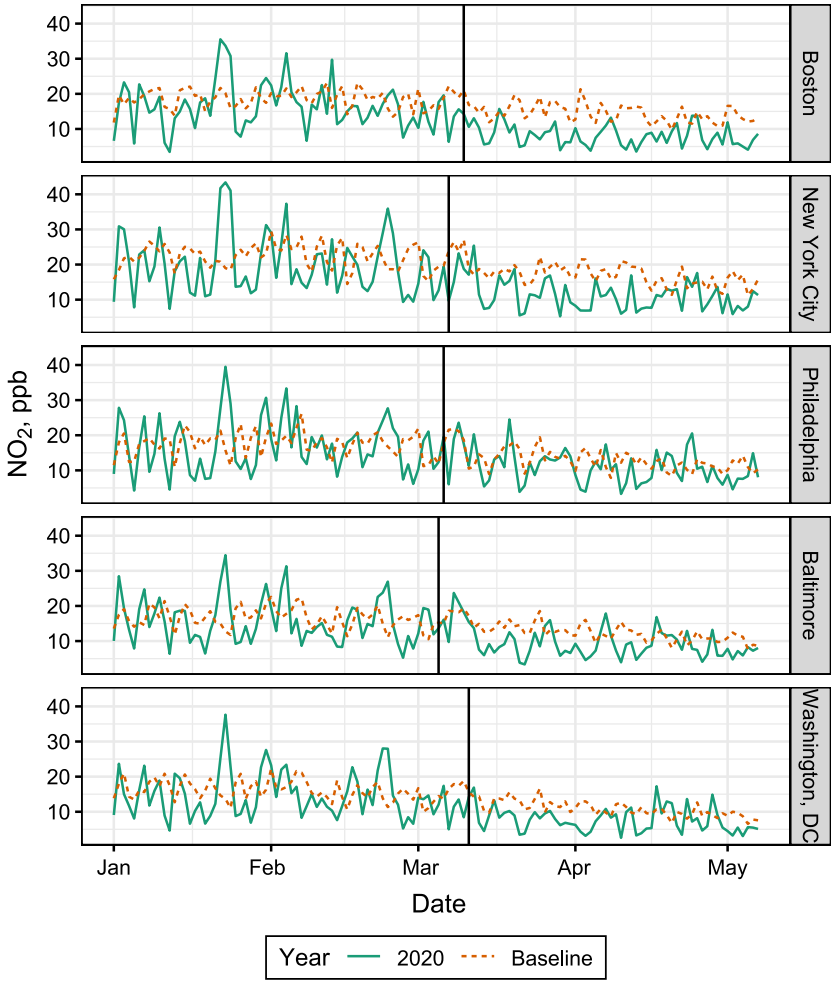

Figure 1. Daily time series of atmospheric nitrogen dioxide (NO_2) concentrations before and during the COVID-19 lockdowns in 5 eastern US cities, 2013-2020. The graph shows time series of daily NO_2 concentrations (parts per billion [ppb]) in each of the 5 cities considered. The solid line indicates the daily concentrations in 2020. The dashed line is a pointwise average of the daily concentrations during the baseline years. The black vertical lines denote the date of the lockdown in each state.

The ATTs from all methods, in 2020 and baseline years, are displayed in Figure 2 (exact values and *P* values or 95% CIs are shown in Table S3). In 2020, most ATTs are negative. The ATTs from GB and RF are similar. Philadelphia and Baltimore consistently have the smallest magnitudes of ATTs in 2020 across methods. The lockdown effect estimated by GB and RF in Philadelphia is almost zero.

Figure 2 also shows which methods identify large effects in baseline years. The mean difference and linear CITS methods yield many effect sizes in baseline years that are larger than in 2020. DiD is the best performing of the non–machine-learning methods, with relatively few large ATT magnitudes estimated in baseline years. The machine learning methods (GB, RF) do very well in baseline years, estimating ATTs with very small magnitudes, while identifying negative ATTs in 2020, except in Philadelphia.

These results show that while mean differences and linear CITS models have been widely deployed to understand the effect of COVID-19 lockdowns on air pollution, they identify many large effects in baseline years owing to reliance on incorrect assumptions (see Methods section, "Conceptual validation"). They cannot reliably estimate the ATT in 2020, despite reporting large decreases in NO_2 in all cities. This is most clearly illustrated in Washington, DC, where large negative ATTs in baseline years are estimated by all methods except GB and RF. Our empirical validation strategy (see Methods, "Empirical validation") highlights a strength of the machine learning methods as compared with mean difference or linear CITS methods, leading us to prefer them for the final conclusions from this case study. For mean difference and linear CITS methods, we also assess the significance of the estimated ATTs in baseline years as an empirical validation technique. Figure S1 shows that these methods produce false-positive results in baseline years. This is further discussed in Appendix S7.

The ATT has the same unit as NO_2 concentrations (parts per billion [ppb]), which is not easily comparable to other pollutants, or even between cities where average concentrations differ. To comparatively assess the machine learning CITS ATTs in 2020, we present them as percent differences using equation (2) (Figure 3 and Table S4), along with 95% CIs. These intervals are obtained using block bootstrapping with a block length of 1 week. We select this window because it yields conservative intervals (Figures S2 and S3) and from looking at the order of autoregression of the residuals (Figure S4). Further discussion of the block bootstrap and uncertainty estimation for machine learning CITS can be found in Appendix S7. We see that the percent decrease in NO_2 concentrations is greatest in Washington, DC (24%-35%). Philadelphia has the smallest observed percent decrease (2%-5%). However, our

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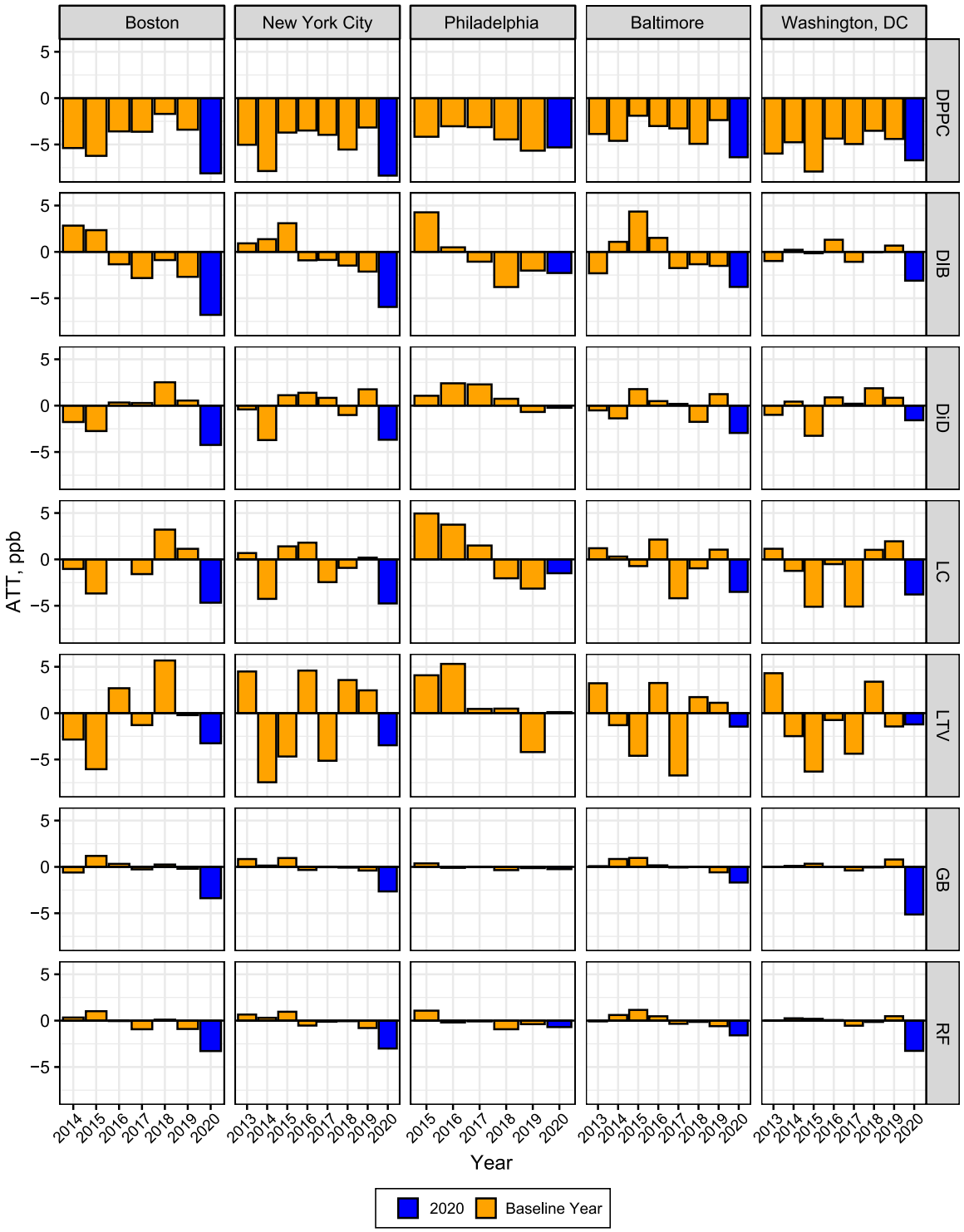

Figure 2. Average treatment effect on the treated (ATT) of NO_2 estimated from mean difference, linear CITS, and machine learning CITS methods in 2020 and baseline years. The graph shows a bar plot of ATT estimates (parts per billion [ppb]) derived by all methods in all years and all cities. The exact values of the ATT and the corresponding *P* values or 95% CIs are included in Table S3. DPPC, difference between the mean values in the postcutoff and precutoff periods; DIB, difference between the mean values for the postcutoff period in the intervention year and the baseline years; DiD, difference in differences; LC, linear model with a constant lockdown effect; LTV, linear model with a time-varying lockdown effect; GB, gradient boosting; RF, random forest.

Philadelphia results may be affected by the monitor locations in the city (Figure S10).

Model validation is also performed to ensure that our proposed machine learning CITS methods are appropriate for the data. We check the predictive power of the linear and machine learning CITS models by predicting on holdout data and comparing these predictions with the observed concentrations. Figures S11-S15 show that the fits from the machine learning models are reasonable for the data. R^2 values, as well as residual plots, are also presented (Figures S16-S22).

We also perform 2 sensitivity tests. The first is to remove all measurements from one of the monitors within a city and

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Figure 3. Percent difference in atmospheric nitrogen dioxide (NO_2) concentrations in the presence and absence of the 2020 COVID-19 lockdowns in 5 eastern US cities as estimated by our machine learning CITS method. The graph shows the percent difference in NO_2 concentrations (error bars, bootstrapped 95% CIs) as estimated by machine learning methods. Percent difference is the average treatment effect on the treated divided by the average counterfactual concentration in the postcutoff period in 2020. GB, gradient boosting; RF, random forest.

estimate the ATT from the remaining monitors, to check the sensitivity of the results to the measurement locations. We also repeat the analysis using only a subset (2017-2019) of the baseline years to check the importance of the baseline period. The machine learning CITS models are robust to both changes (Figure 4, Figures S23 and S24).

A simulation study is also used to investigate the methods' ability to accurately estimate the true intervention effect. We see in Figures S25 and S26 that the machine learning CITS models estimate the ATT well under both linear and nonlinear data-generation mechanisms, while the linear and mean difference methods have larger biases when the data generation is nonlinear. Further details of the validation analyses are presented in Appendix S10.

Discussion

Review of approaches to estimate causal effects in time series data

We have presented a flexible machine learning CITS approach for robust estimation of the causal impact of a policy change or natural experiment potentially resulting in a perturbation in air quality. We discuss the similarities and differences of our method with other approaches in the literature. Machine learning has been used to estimate the effect of COVID-19 lockdowns without the CITS framework,[16] which does not account for the data noise and can be biased due to model misspecification. Other approaches with which to estimate causal effects in time series are synthetic controls,[40] propensity scores for ITS,[41] structural

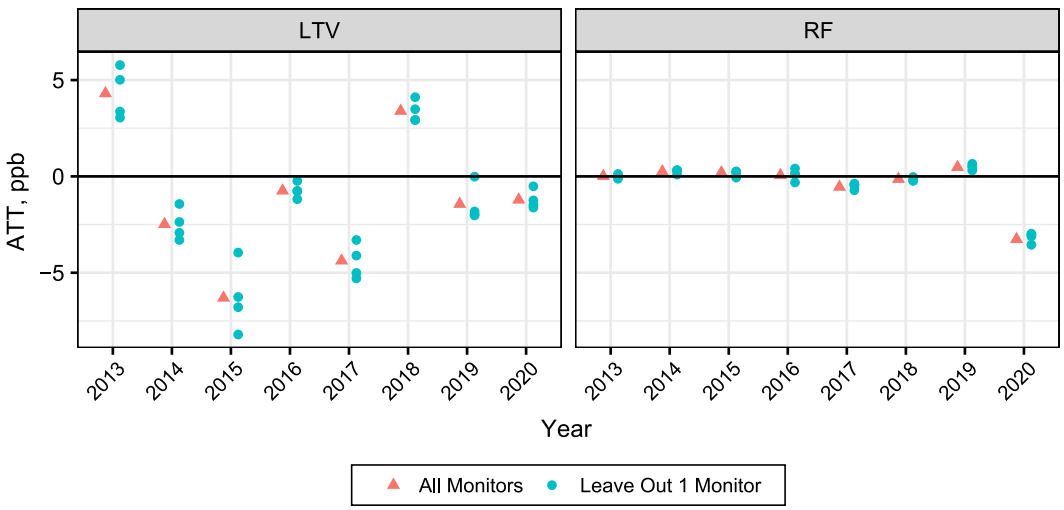

Figure 4. Average treatment effect on the treated (ATT) estimated from a sensitivity analysis in Washington, DC, where one monitor at a time is left out, compared with the full analysis using all monitors. Results for 1 linear model and 1 machine learning model are shown; the full results for all methods and all cities are shown in Figure S23. Red triangles show the results with all monitors included in the analysis; blue circles indicate the estimated ATTs when leaving out one of the monitors in the city. LTV, linear model with a time-varying lockdown effect; ppb, parts per billion; RF, random forest.

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Table 2. Comparison (%[a]) of our machine learning CITS method (with bootstrapped 95% CIs) with those of other studies. Changes in NO_2 in 4 eastern US cities due to the COVID-19 pandemic in 2020 are presented.

City and state	Study and method			
	This study		Other studies	
	Machine learning CITS fitted using gradient boosting	Machine learning CITS fitted using random forest	Cooper et al[46]: comparison of mean in 2020 with mean in 2019, with adjustments for meteorological factors and long-term trends	Goldberg et al[47]: comparison of mean in 2020 with mean in 2019, normalization by standard meteorological factors, and chemical transport model
Boston, Massachusetts	−26.00 (−34.89 to −17.12)	−23.44 (−30.82 to −16.06)	−6.74	−22.8
New York, New York	−16.40 (−25.47 to −7.33)	−17.22 (−24.26 to −10.19)	−10.6	−20.0
Philadelphia, Pennsylvania[b]	−1.93 (−5.34 to 1.49)	−5.17 (−11.33 to 0.98)	−11.29	−30.7
Washington, DC	−34.72 (−55.92 to −13.52)	−24.05 (−32.94 to −15.16)	−20.33	−21.2

Abbreviations: CITS, comparative interrupted time series; NO_2, nitrogen dioxide.
[a]Percent difference in atmospheric NO_2 concentrations before and during the 2020 COVID-19 lockdowns in 4 eastern US cities.
[b]In the dataset used for this analysis, there were no sites measuring NO_2 levels in Philadelphia near the downtown area. See Appendix S12 for more discussion.

time series,[42] and doubly robust methods.[43,44] The main limitation of synthetic controls for our case study is that with few control groups (7 baseline years in our analysis) and many time points and covariates, satisfying the synthetic control equations is not possible. Propensity scores are not applicable here because lockdowns cannot be predicted using meteorological factors. Details of these approaches' limitations are shown in Appendix S11.

Our validation framework (see Methods section) is the second main contribution of this paper. Haber et al[45] also discuss the validity of ITS, CITS, and DiD studies calculating effects of a COVID-19 policy on outcomes, based on the functional form, threshold choice, and consideration of concurrent changes. We offer validation approaches specific to the context of air pollution. We state the necessary assumptions for causal identification and demonstrate how common methods can be framed as CITS but rely on additional strong assumptions. We also offer a simple empirical diagnostic to rule out methods yielding large effects in baseline years.

Conclusions

Our proposal to use machine learning for CITS produces a causal estimate under certain assumptions (see "Conceptual validation" in Methods), adjusts for meteorological factors and other covariates, and does not rely on parametric assumptions like linearity. Our approach found a decrease in NO_2 concentrations in 4 US cities during the COVID-19 lockdowns. Empirical validation revealed that many common methods like mean differences and linear models identified large effects in baseline years. Only the machine learning CITS methods did not estimate large effects in baseline years.

In Appendix S12 and Table S5, we present a comparison of our method with methods used for published estimates of the lockdown effect on NO_2 levels. We find that many papers do not incorporate meteorological factors into their estimates, and almost none check for false effects as our validation framework does. Our ATT estimates broadly agree with those of Cooper et al[46] and Goldberg et al,[47] who also accounted for meteorological factors and trends (Table 2).

Our approach is general and can be used widely, across other regions, pollutants, and sharp interventions. Such studies have long existed[23-26] and will likely become increasingly common as air pollution data become more accessible and spatiotemporally resolved. This article provides a timely assessment of a suite of common models for such analyses and proposes new machine learning CITS methods that are less assumption-driven and more data-driven. Not all studies require machine learning, and simpler models may be justified if, conceptually, the additional assumptions (see "Conceptual validation" above) are plausible and, empirically, the methods do not result in large effects during baseline periods (see "Empirical validation"). Additionally, NO_2 data are high-quality, but in other applications investigators must determine whether data quality may change over time, such as due to changes in instrumentation, calibration, or source emissions. Other studies may also require further consideration of the role of synoptic/regional weather patterns, though our results showed that using city-level covariates was sufficient to adequately model our data.

Machine learning methods are not without issues. They can overfit if regularization is inadequate. Additionally, uncertainty quantification of ATT estimates from machine learning methods on time-series data remains challenging. Developing a principled methodology for this is beyond the scope of this paper but is an important future direction. Additionally, not all machine learning methods will work well for a particular problem, so it is important to compare the performance of different machine learning models to select an appropriate method. Our validation framework can be used to select a preferred method in such studies.

Supplementary material

Supplementary material is available at *American Journal of Epidemiology* online.

Funding

This work was supported by the US Environmental Protection Agency (assistance agreement RD835871 with Yale University); the Bloomberg American Health Initiative at the Johns Hopkins Bloomberg School of Public Health, in collaboration with the Alliance for a Healthier World at Johns Hopkins University; the Fonds de recherche du Québec—Nature et Technologies (bourse de maîtrise B1X to C.H.); the National Science Foundation Graduate Research Fellowship Program (grant DGE2139757 to C.H. and grant DGE1752134 to C.B.); and the National Institute of Environmental Health Sciences (grant R01 ES033739 to C.H., K.K., R.D.P., and A.D. and awards K99ES029116 and R00ES029116 to M.L.Z.).

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Conflict of interest

The authors have no potential conflicts to disclose.

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Data availability

The data and software code used in this study are available at https://github.com/cmheffernan/ML_CITS_air_pollution.git.

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American Journal of Epidemiology, 2025, **194**, 185–194
https://doi.org/10.1093/aje/kwae171
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